

A-level

ENVIRONMENTAL SCIENCE

Questions by topic: Soils: fertility, erosion & management - Set 1

Mark scheme

AQA spec 3.2.5; The physical environment; textbook pp. 158-169

Marking guidance

- The number in brackets after each marking point is the mark it earns; the note above the points says how many are needed.
- Where a question asks for a fixed number of points (eg *two reasons*), credit only the first points given.
- Alternative correct wording and other relevant, accurate points are credited.
- For levels-of-response questions, first determine the level using the descriptors, then position the mark within the level using the indicative content.
- Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	<i>1 mark for each point.</i> • B (about 40:40:20 sand:silt:clay) (1) • Balanced mix gives good drainage and aeration from the sand fraction (1) • Water retention and nutrient (ion) retention from the clay fraction; easy root penetration and cultivation (1)	3
01.2	<i>1 mark for each point.</i> • Large sand particles leave large pore spaces so water drains rapidly (high permeability / aeration) (1) • Nutrient ions adsorb onto the (charged) surfaces of clay particles but not onto sand (1) • With only 5% clay there is little surface to hold ions so nutrients are leached; low water retention too (1)	3
01.3	<i>1 mark for each point.</i> • C (clay) holds much water so has a high thermal capacity: warms and cools slowly; A warms rapidly after cold weather (1) • Clay particles held together by capillary water so roots find it hard to push between them; penetration is easy in sandy A (1)	2
01.4	<i>1 mark for each point.</i> • Shake a known mass of soil with water in a measuring cylinder and leave to settle; particles settle in order of size (sand first, then silt, then clay; organic matter floats) (1) • Measure the depth / volume of each layer to calculate the percentage of each fraction, then use the soil triangle to name the textural class (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 158-160, 412; AQA spec 3.2.5.1.	

Question	Marking guidance	Marks
02.1	<i>Up to 6 marks in total; award marks in proportion to the completeness of each point.</i> • Detritivores (worms, beetle larvae, millipedes, woodlice) physically break up dead organic matter, increasing its surface area, and release nutrients; worms create tunnels improving drainage and aeration (up to 2) • Decomposers (bacteria, fungi) secrete enzymes that break down dead organic matter, releasing nutrient ions (ammonification) and forming humus which binds particles into crumb peds (up to 2) • Nitrogen-fixing bacteria convert nitrogen gas to ammonium ions; nitrifying bacteria oxidise ammonium to nitrite then nitrate for root absorption (up to 2) • Mycorrhizal fungi form symbiotic associations with roots increasing phosphate uptake (up to 2)	6
02.2	<i>1 mark for each point.</i> • This is the range of tolerance of most plants and soil biota; acidic soils leach nutrients and mobilise toxic aluminium, damaging root membranes (1) • In alkaline soils phosphates become insoluble / unavailable (1)	2
02.3	<i>1 mark for each point.</i> • Interstitial spaces are reduced so aeration falls (aerobic processes such as nitrification and decomposition slow) and roots cannot penetrate (1) • Infiltration falls so waterlogging is more likely and runoff / erosion increase (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 158-161; AQA spec 3.2.5.1.	
03.1	<i>1 mark for each point.</i> • Soil loss increases as vegetation cover decreases: forest lowest (0.5), bare soil highest ($60 \text{ t ha}^{-1} \text{ yr}^{-1}$) - over 100 times more (1) • Ploughing downslope loses nearly three times as much as contour ploughing for the same crop (1)	2
03.2	<i>Any 4 from the points below - 1 mark each, maximum 4.</i> • Leaves / leaf litter intercept raindrops so soil particles are not dislodged by rain splash (1) • Roots bind soil particles together (1) • Vegetation acts as a windbreak reducing wind velocity and the kinetic energy available to carry particles (1) • Plants increase infiltration (root channels, organic matter) so less surface runoff (1) • Organic matter / humus from vegetation binds particles into stable peds (1)	4
03.3	<i>1 mark for each point.</i> • Furrows running downslope act as channels / gulleys that increase the velocity (kinetic energy) of runoff (1) • Furrows along the contour trap the water so it loses energy, deposits particles and infiltrates (1)	2
03.4	<i>1 mark for each point.</i> • In the dry season the soil (especially low-clay soils) is loose and dry with little cohesion so wind blows it away (1) • In the wet season heavy rain causes rain splash and surface runoff when infiltration capacity is exceeded (1)	2

Question	Marking guidance	Marks
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 162-164; AQA spec 3.2.5.2.	
04.1	<i>2 marks for each point (partial credit for a partly correct point).</i> • Overgrazing: vegetation is eaten faster than it grows so soil is exposed; hooves disturb the soil, damage roots and compact it, reducing infiltration and increasing runoff (2) • Reduced biota (ploughing, agrochemicals, less organic matter): less humus so less adhesion between particles; fewer worms so less aeration and lower infiltration capacity; fewer nutrients released so less vegetation cover (2)	4
04.2	<i>1 mark for each point.</i> • Tree roots that held the soil together decompose so the soil is bound less strongly (1) • Heavy rain adds mass and lubricates particles; without interception more water enters the soil, making slumping more likely (1)	2
04.3	<i>Any 4 from the points below - 1 mark each, maximum 4.</i> • Sedimentation in rivers reduces flow capacity so flooding downstream is more likely (1) • Sedimentation in reservoirs reduces their storage volume (1) • Coastal sedimentation covering reefs / seagrass; turbidity reducing light for aquatic plants (1) • Increased atmospheric particulates from wind erosion worsening asthma; dust covering crops / urban areas (1) • Desertification as vegetation cannot re-establish; reduced rainfall (1) • Nutrients (phosphates) carried on soil particles causing eutrophication (1)	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 162-164; AQA spec 3.2.5.2.	
05.1	<i>A correct final answer (with unit where asked) scores full marks. If the answer is wrong, award 1 mark for each correct step of working shown.</i> • $110 \times 0.67 \times 1.35 \times (0.50 \times 0.90) \times 1.0$ (1) • $= 44.8 \text{ t ha}^{-1} \text{ yr}^{-1}$ (accept 44.7-44.8) (1)	2
05.2	<i>A correct final answer (with unit where asked) scores full marks. If the answer is wrong, award 1 mark for each correct step of working shown.</i> • $110 \times 0.67 \times 1.35 \times (0.35 \times 0.60) \times 0.25$ (1) • $= 5.2 \text{ t ha}^{-1} \text{ yr}^{-1}$ (1)	2
05.3	<i>1 mark for each point.</i> • Beans / up-and-down ploughing: severe class (over 30) and about 45 times the formation rate - clearly unsustainable, soil depth will fall (1) • Wheat / mulch / strip cropping: moderate class, reduced by about 88% but still 5 times the formation rate so not fully sustainable (1) • Further measures (long-term crops, terracing, windbreaks, more organic matter) or grazing land ($C = 0.02$) needed to bring loss below $1 \text{ t ha}^{-1} \text{ yr}^{-1}$ (1)	3
05.4	<i>1 mark for each point.</i> • Factors are averages / arbitrary values for typical conditions; rainfall erosivity varies between years / storms (1) • Does not include gully erosion, landslides, wind erosion or local variation in soil; based on data from particular regions (1)	2

Question	Marking guidance	Marks
05.5	<i>1 mark.</i> • C (crop / tillage) or P (erosion control practice) - R, K and LS are fixed by climate, soil and topography (1)	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 167-169; AQA spec 3.2.5.2.	
06.1	<i>1 mark for each point.</i> • Terracing: sloping land divided into narrow level fields held by retaining walls along contours; water slows as it crosses each terrace and infiltrates instead of running off (1) • Tied ridging: on nearly flat land a criss-cross of intersecting ridges retains rainwater, increasing infiltration and preventing runoff (1) • Windbreaks: hedgerows / tree rows reduce wind velocity (for up to 30 times their height) so the wind has less kinetic energy to carry particles; protect bare fields after harvest (1) • Mulching: organic matter on the surface protects soil from raindrop impact and wind; decomposes to humus which binds particles; retains moisture (1) • Strip cropping: strips of different crops harvested at different times (at 90° to the wind) so there is always a crop protecting part of the field; slows runoff (1)	5
06.2	<i>1 mark for each point.</i> • No frequent replanting / soil disturbance (ploughing) that breaks up soil structure and exposes particles (1) • Permanent root systems and canopy cover protect the soil all year (1)	2
06.3	<i>Any 3 from the points below - 1 mark each, maximum 3.</i> • Control stocking density so vegetation regrows faster than it is eaten (1) • Restrict access to vulnerable areas such as river banks (fencing) (1) • Move / remove livestock at high-risk times, eg after heavy rain when soil is wet (1) • Rotational grazing to rest pastures (1)	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 164-166; AQA spec 3.2.5.3.	
07.1	<i>A correct final answer (with unit where asked) scores full marks. If the answer is wrong, award 1 mark for each correct step of working shown.</i> • Water = $65.0 - 55.0 = 10.0$ g; fresh soil = 40.0 g (1) • $10.0 / 40.0 \times 100 = 25.0\%$ (1)	2
07.2	<i>A correct final answer (with unit where asked) scores full marks. If the answer is wrong, award 1 mark for each correct step of working shown.</i> • Organic matter = $55.0 - 52.6 = 2.4$ g; dry soil = 30.0 g (1) • $2.4 / 30.0 \times 100 = 8.0\%$ (1)	2
07.3	<i>1 mark for each point.</i> • So all water is removed and the mass loss on burning is due to organic matter only (1) • Heating until mass no longer changes shows all water has evaporated (1)	2

Question	Marking guidance	Marks
07.4	<i>1 mark for each point.</i> - Organic matter releases nutrients as it decomposes and provides food for soil biota (1) - Increases water retention and, as humus, binds particles into crumb peds improving structure / aeration (1) - Surface organic matter / humus adhesion reduce rain splash and wind erosion (1)	3
07.5	<i>1 mark.</i> Samples taken from different depths / locations (organic matter is concentrated near the surface); incomplete burning; different heating times (1)	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 158-159, 412; AQA spec 3.2.5.1, 3.7.2.1.	
08.1	<i>Correct answer only.</i> B - tiny pores between clay particles allow capillary rise; none in sand (1)	1
08.2	<i>1 mark for each point.</i> - Hygroscopic: held tightly by adhesive forces on particle surfaces; cannot drain and cannot be absorbed by plants (1) - Capillary: held in pores by surface tension; does not drain but can be absorbed by plants (1) - Gravitational: drains freely through large pores; available to plants only briefly as it passes (1)	3
08.3	<i>1 mark.</i> Small round crumbs leave spaces giving good drainage, aeration and easy root penetration; large flat platy peds pack together reducing all three (1)	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 158-169; AQA spec 3.2.5.	